

CNC PLASMA

QUICK CUT FLOW CHART

A3 LANDSCAPE | TRAINED TEACHER REFERENCE

Read down each column, then move to the next.  
Each numbered box links to the full operating guide.

START GATE - STEP 01 MUST ALREADY BE COMPLETE

Trained, authorised teacher; supplier-checked installation and completed setup record beside the screen. This chart recalls taught steps. It does not supply unverified recipe values, switch locations or permission to operate.

Already have a checked TAP? Skip A only when its file revision, material and recorded CAM setup are confirmed.

A PREPARE THE FILE

Computer only | USB DXF to TAP

02 Open and size-check the DXF

SheetCam: File > Import drawing

Select USB > your DXF > Open. Choose Metric + bottom-left drawing position > OK. Measure: square 100 x 100 mm; centre 50,50 mm; hole diameter matches drawing.



03 Create and preview the cut paths

Operations > jet-cutting icon

Outside Offset; correct Layer and recorded Tool, Path rules and lead-ins > OK. Show tool paths, starts and rapids. Run simulation: hole first, square last; leads in waste; no missing or extra cuts.



04 Save the machine file

File > Save job as; Run post processor

Use the supplier-confirmed post-processor. Save the checked revision as .tap. Copy that TAP from USB into the recorded SureControl folder. Keep DXF and editable job separately.

Continue at B (top of next column) >

B PREPARE THE MACHINE

Authorised teacher | real equipment

05 Secure the area

Students out | guards + extraction

Dry, clear area; flammables removed. Check guards, slag drawer, screens, curtains and interlocks. Confirm screened operator position. Keep equipment isolated for access/loading.



06 Match sheet, torch and power

Use the completed machine setup record

Support sheet; check work-return clamp + CPC, consumables, air and water if fitted. Follow recorded power-on order; extraction running. Set PHYSICAL source current to match the job recipe.



07 Connect SureControl

Connect > Offline > Online

Open SureControl: Disconnected. Click Connect; expect Offline (button becomes Reset). With travel clear, select Online. Fresh start: expect Not Referenced.



08 Reference the axes

Reference > recorded axes > Start > OK

Watch from the designated position. Wait for normal completion before OK. This moves the machine; it does not set the part origin.

Continue at C (top of next column) >

C LOAD, CHECK, POSITION

Authorised teacher | people clear

09 Load the checked job

Load > folder > correct .tap > OK

Import DXF stays unticked for this TAP route. Check filename/revision, square, hole and orientation. A new job normally shows Set Origin.



10 Verify Touch sensing

Recorded slow jog + sensor test

For the confirmed contact test: Page Down gently to contact; RELEASE. Touch INDICATOR turns green. Page Up to recorded clearance; indication clears. No Shift/full-speed jog.



11 Torch test ONLY if recorded

Torch On/Off icon = REAL FIRING

If required, use recorded test position, protection and duration; extraction/screens active. Click to fire, click again to stop; verify arc out. If explicitly not required, continue. Blank test record: STOP.



12 Position and assign part zero

Arrow keys to jog > Set Origin

Use recorded clearance/speed. Allow room for square, kerf, leads and travel. Release keys; click Set Origin. Check zero/crosshair. Origin is a MOVE command; do not confuse the two.

Continue at D (top of next column) >

D DRY RUN, CUT, FINISH

Authorised teacher | final decision

13 Dry run with NO arc

Simul ON > Start from beginning

Real machine movement. Observe full route: hole first, square last; all travel fits and clears obstructions. No firing. Wait for Idle. Keep Simul ON until ready to cut.



ALL DRY-RUN CHECKS PASS?

YES: continue down. NO: STOP below.



14 Final check, then REAL CUT

Cut Parameters > check recipe > OK

Match file/origin, sheet, consumables and PHYSICAL current. Confirm CAM/controller settings agree. Read the option strip below. Extraction running; access secured; teacher behind screens/curtains. Start from beginning.



15 Finish and isolate before access

Idle > recorded shutdown + isolation

Wait for arc out and motion stopped. Allow recorded cooling/post-flow/extraction timing; verify electrical isolation before student entry. Retrieve with specified tools/PPE when safe; measure and record result.

BEFORE REAL CUT - READ EACH OPTION

Touch ON | Laser OFF | Online ON | Simul OFF | THC = recorded state | Kerf = recorded state (no double compensation)

ANY FAILED CHECK / WRONG STATE / UNSAFE ACCESS: STOP

Do not cut. During danger, use the taught emergency stop. Resolve faults and repeat affected checks/dry run. Stop or Offline is not electrical isolation.